

## TAKADA, Japan

WMO: 476120

Lat: 37.1067N

Lon: 138.2467E

Elev: 48

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>27.3 | (c)<br>29.0 | (d)<br>19.2                   | (e)<br>14.4 | (f)<br>34.4 | (g)<br>21.0 | (h)<br>15.8 | (i)<br>35.0 | (j)<br>19.3           | (k)<br>39.3 | (l)<br>17.1 | (m)<br>38.0 | (n)<br>5.7               | (o)<br>180 | (p)<br>0.386 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>14.3            | (c)<br>92.4     | (d)<br>78.1 | (e)<br>90.2 | (f)<br>77.5 | (g)<br>87.9 | (h)<br>76.7 | (i)<br>79.6         | (j)<br>88.8 | (k)<br>78.5 | (l)<br>87.3 | (m)<br>77.6 | (n)<br>85.6 | (o)<br>7.7           | (p)<br>340 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.1                            | 141.2 | 84.2 | 76.1 | 136.7 | 83.3 | 75.2 | 132.3 | 82.4 | 43.1          | 89.3 | 42.1 | 87.6 | 41.1 | 85.9 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 17.6          | (b) 14.3 | (c) 12.1 | (e) 22.9                   | (f) 97.9 | (g) 2.2            | (h) 2.3 | (i) 21.3                                           | (j) 99.5 | (k) 20.1   | (l) 100.9 | (m) 18.8   | (n) 102.2 | (o) 17.2   | (p) 103.9 |    |    |
| (a) 21.7          | (b) 81.5 | (c) 2.0  | (e) 21.7                   | (f) 81.5 | (g) 2.0            | (h) 1.2 | (i) 20.3                                           | (j) 82.3 | (k) 19.1   | (l) 83.0  | (m) 18.0   | (n) 83.6  | (o) 16.6   | (p) 84.5  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 57.5   | 36.6  | 37.0  | 42.8  | 53.1  | 62.8  | 70.0  | 77.9  | 80.4  | 72.9  | 62.2  | 51.2  | 41.8  |      |
|                                                                                  | DBStd                                                                            | 16.02  | 4.09  | 4.46  | 5.85  | 6.50  | 5.50  | 4.45  | 4.51  | 4.28  | 5.46  | 5.38  | 5.74  | 5.30  |      |
|                                                                                  | HDD50                                                                            | 1373   | 415   | 366   | 238   | 40    | 0     | 0     | 0     | 0     | 0     | 1     | 53    | 260   |      |
|                                                                                  | HDD65                                                                            | 4086   | 880   | 785   | 689   | 359   | 108   | 9     | 0     | 0     | 3     | 120   | 414   | 719   |      |
|                                                                                  | CDD50                                                                            | 4113   | 0     | 1     | 13    | 134   | 397   | 599   | 865   | 942   | 687   | 379   | 90    | 6     |      |
|                                                                                  | CDD65                                                                            | 1351   | 0     | 0     | 0     | 3     | 40    | 158   | 400   | 477   | 240   | 33    | 0     | 0     |      |
|                                                                                  | CDH74                                                                            | 10860  | 0     | 0     | 1     | 43    | 297   | 774   | 3328  | 4735  | 1548  | 133   | 1     | 0     |      |
|                                                                                  | CDH80                                                                            | 3736   | 0     | 0     | 0     | 5     | 40    | 135   | 1186  | 1897  | 453   | 20    | 0     | 0     |      |
| Wind                                                                             | WSAvg                                                                            | 5.1    | 5.9   | 5.5   | 5.6   | 5.8   | 5.3   | 4.4   | 4.1   | 4.2   | 4.3   | 4.7   | 5.1   | 5.9   |      |
| Precipitation                                                                    | PrecAvg                                                                          | 99.1   | 14.4  | 9.1   | 6.9   | 3.9   | 3.4   | 5.4   | 7.6   | 6.3   | 7.5   | 7.6   | 11.3  | 15.9  |      |
|                                                                                  | PrecMax                                                                          | 118.4  | 21.5  | 14.3  | 10.7  | 6.4   | 7.3   | 10.2  | 18.3  | 16.8  | 11.3  | 14.3  | 20.1  | 29.5  |      |
|                                                                                  | PrecMin                                                                          | 79.2   | 7.6   | 3.3   | 3.8   | 0.8   | 1.5   | 2.0   | 1.1   | 0.9   | 1.9   | 4.3   | 4.4   | 6.5   |      |
|                                                                                  | PrecStd                                                                          | 11.5   | 3.6   | 2.8   | 1.7   | 1.4   | 1.5   | 2.2   | 4.1   | 3.5   | 2.7   | 2.3   | 3.5   | 5.1   |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 54.6  | 59.0  | 70.2  | 78.9  | 83.8  | 87.4  | 94.3  | 96.9  | 92.1  | 82.5  | 71.9  | 62.8  |      |
|                                                                                  |                                                                                  | MCWB   | 45.5  | 50.9  | 53.9  | 60.2  | 64.5  | 72.5  | 77.8  | 78.2  | 76.3  | 69.8  | 59.0  | 53.0  |      |
|                                                                                  | 2%                                                                               | DB     | 49.2  | 52.2  | 62.6  | 73.7  | 80.6  | 83.5  | 91.5  | 93.3  | 88.6  | 77.3  | 67.5  | 57.5  |      |
|                                                                                  |                                                                                  | MCWB   | 42.3  | 44.1  | 49.4  | 57.8  | 63.8  | 70.8  | 77.9  | 78.5  | 75.8  | 66.2  | 57.4  | 49.2  |      |
|                                                                                  | 5%                                                                               | DB     | 46.1  | 48.4  | 57.9  | 69.3  | 77.3  | 81.0  | 89.2  | 91.3  | 85.2  | 74.2  | 64.2  | 53.8  |      |
|                                                                                  |                                                                                  | MCWB   | 40.6  | 41.4  | 47.9  | 55.3  | 62.5  | 69.5  | 77.4  | 78.2  | 74.5  | 64.9  | 55.8  | 47.4  |      |
|                                                                                  | 10%                                                                              | DB     | 43.4  | 45.1  | 53.6  | 65.4  | 74.2  | 78.5  | 86.7  | 89.1  | 81.9  | 71.5  | 61.2  | 50.7  |      |
|                                                                                  |                                                                                  | MCWB   | 38.9  | 39.7  | 45.9  | 54.0  | 61.5  | 68.6  | 76.4  | 77.6  | 72.7  | 63.0  | 54.5  | 45.5  |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 47.0  | 50.0  | 55.0  | 62.8  | 68.9  | 75.9  | 80.7  | 81.4  | 78.9  | 71.8  | 62.0  | 54.7 |
|                                                                                  |                                                                                  |        | MCDB  | 52.8  | 57.9  | 66.1  | 72.2  | 77.1  | 83.1  | 90.4  | 90.8  | 88.2  | 79.1  | 67.0  | 58.8 |
| 2%                                                                               |                                                                                  | WB     | 43.4  | 45.2  | 51.5  | 59.6  | 66.5  | 73.6  | 79.2  | 79.9  | 76.9  | 68.9  | 59.4  | 50.7  |      |
|                                                                                  |                                                                                  | MCDB   | 47.7  | 50.2  | 59.9  | 69.6  | 75.4  | 79.2  | 88.3  | 89.4  | 85.4  | 74.3  | 64.4  | 55.9  |      |
| 5%                                                                               |                                                                                  | WB     | 41.3  | 42.4  | 49.0  | 57.4  | 64.7  | 72.0  | 78.1  | 79.0  | 75.5  | 66.8  | 57.6  | 48.0  |      |
|                                                                                  |                                                                                  | MCDB   | 45.1  | 47.1  | 55.8  | 66.1  | 73.0  | 77.6  | 86.6  | 88.1  | 82.8  | 71.8  | 62.4  | 52.8  |      |
| 10%                                                                              |                                                                                  | WB     | 39.3  | 40.0  | 46.4  | 55.3  | 63.0  | 70.4  | 77.0  | 78.2  | 74.1  | 64.8  | 55.4  | 45.9  |      |
|                                                                                  |                                                                                  | MCDB   | 43.0  | 44.6  | 53.4  | 63.2  | 71.3  | 76.2  | 84.7  | 86.6  | 80.3  | 69.7  | 60.2  | 50.2  |      |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB                                                                            | MDBR   | 10.1  | 12.2  | 15.7  | 18.8  | 18.2  | 14.5  | 13.1  | 14.3  | 14.2  | 15.5  | 15.1  | 12.1  |      |
|                                                                                  |                                                                                  | MCDBR  | 16.0  | 19.2  | 24.9  | 26.1  | 24.0  | 18.6  | 17.3  | 17.4  | 17.7  | 18.3  | 20.1  | 17.9  |      |
|                                                                                  |                                                                                  | MCWBR  | 10.8  | 12.7  | 14.9  | 13.1  | 11.0  | 8.6   | 7.1   | 7.0   | 7.8   | 9.5   | 11.9  | 11.7  |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 15.0  | 17.8  | 22.2  | 21.7  | 19.3  | 14.5  | 15.5  | 15.9  | 15.6  | 14.9  | 17.0  | 16.4  |      |
|                                                                                  |                                                                                  | MCWBR  | 10.9  | 12.6  | 14.2  | 11.9  | 10.1  | 7.4   | 6.9   | 6.9   | 7.3   | 8.5   | 11.2  | 11.5  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             |        | 0.408 | 0.444 | 0.505 | 0.542 | 0.555 | 0.579 | 0.574 | 0.564 | 0.499 | 0.453 | 0.422 | 0.412 |      |
|                                                                                  | taud                                                                             |        | 1.970 | 1.886 | 1.826 | 1.839 | 1.872 | 1.887 | 1.961 | 1.991 | 2.123 | 2.148 | 2.142 | 2.037 |      |
|                                                                                  | Ebn at Noon                                                                      |        | 232   | 240   | 239   | 239   | 239   | 232   | 232   | 231   | 239   | 238   | 229   | 221   |      |
|                                                                                  | Edh at Noon                                                                      |        | 43    | 53    | 62    | 65    | 64    | 63    | 58    | 55    | 46    | 41    | 36    | 38    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           |        | 545   | 800   | 1151  | 1479  | 1690  | 1603  | 1520  | 1562  | 1230  | 965   | 656   | 488   |      |
|                                                                                  | RadStd                                                                           |        | 44    | 73    | 91    | 123   | 174   | 114   | 226   | 161   | 95    | 82    | 52    | 55    |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (25 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page